PRD-001 AI Hub Prototype: Stock AI and Operations AI

Oct 8, 2026 · @Rajat

Product Requirements Document · Stock AI and Operations AI on the Ellementry data warehouse · Release 1: rule-based agents, dashboards and the data API

Document control

Item	Value
PRD ID	PRD-001
Status	Draft v0.1, for review
Owner	@Rajat
Approver	To be named (open decision OD-08)
Release covered	Prototype Release 1: data API, Stock and Operations dashboards, rule-based Stock AI and Operations AI writing to the insight schema
Builds on	Data warehouse v2.2 (schema v2.0 + migrations to 105, transform v2.2), all 141 tests passing
Design references	How the Schema Works · Target Data Model v2 · Data Context & Business Rules Register · AI Hub Architecture · Our Data Foundation · Data Issues and Doubts Log (93 items)

ID conventions used throughout: business rules BR-nnn and gaps G-nnn as stored in `meta`; functional requirements FR-STK-nn (Stock AI), FR-OPS-nn (Operations AI), FR-API-nn, FR-DSH-nn, FR-INS-nn (insight and decisions); non-functional NFR-nn; open decisions OD-nn; test cases TC-xx-nn; engineering tasks AIH-nnn. Anything marked **PROPOSED** is a default the author suggests and the business must confirm before go-live; nothing marked PROPOSED is treated as a confirmed rule.

Version	Date	Change
0.1	8 Oct 2026	First draft from the v2.2 build, the registers and the issue log

1. Business Requirements

Ellementry needs one trusted answer to two daily questions: where is our stock and what should we do about it, and where are goods stuck on their way in, out or back. Today these answers are pieced together by hand from 14 kinds of Excel export (ERP, Blinkit, Instamart, product master), each with its own traps: GST included under misleading names, the channel hidden in a contact name, locations spelled differently in every file. Stock sent to franchise, SIS and quick-commerce partners vanishes from view even though it is still ours (BR-022), so cover looks shorter or longer than it is and reorders are guessed.

The data warehouse (v2.2) now turns those exports into checked sales, returns, transfers, purchase and stock figures. This PRD covers the next step: putting that data in front of the people who decide, and adding the first two agents. Release 1 agents are rule-based: every number and every recommendation can be traced to a rule, a row and a file. Language-model reasoning, forecasting models and the remaining seven agents are later releases (section 12).

1.1 Business Objectives

1. **Stock visibility everywhere.** Every sellable unit is visible by SKU and location, including Amazon FBA warehouses, own stores, Blinkit and Instamart dark stores, partner consignment and stock with employees, each location counted once (BR-025).
2. **Fewer stockouts and less dead stock.** The business receives reorder, Bagru-to-FBA transfer and discontinued-stock recommendations based on real demand, real stock and real supplier lead times, and records a decision on each.
3. **Operational exceptions surfaced early.** Transfers stuck in transit, overdue purchase orders, unusual return volumes and, once the dispatch export arrives, late shipments and RTOs are raised as alerts with an owner instead of being found weeks later.
4. **Numbers the team trusts.** Dashboard figures reconcile to the source exports, gaps show as "no data yet" rather than zero, and the team validates the numbers before agents are relied on.

5. **A measurable record of agent advice.** Every recommendation and alert is stored with who accepted or rejected it and when (BR-028), so the agents' usefulness can be measured before they are trusted further.

1.2 Non-goals for This Release

Statistical or machine-learning demand forecasting, LangGraph orchestration and a conversational (LLM) interface are out of scope; Release 1 uses transparent rules. The other seven agents (Executive, Marketing, Commerce, Product, Finance, Customer, Design) are out of scope. Automated ingestion (n8n, scheduled pulls from seller panels) is out of scope: files are loaded by hand with the existing loader. Alert delivery to Slack, WhatsApp or email is out of scope (alerts live in the app; OD-05). Writing back to the ERP (creating POs or transfer orders) is out of scope: agents recommend, people act in the ERP. Cloud deployment is out of scope; Release 1 runs on a laptop with local PostgreSQL, keeping SQL portable. Replenishment of Blinkit and Instamart dark stores is out of scope because the supply route is unresolved (issue D-092); their stock and cover are reported but not recommended on.

1.3 Success Criteria

- The warehouse builds from a clean unzip on the owner's laptop with all 141 existing tests passing and the verified control totals reproduced: retail net sales Rs 4,22,80,208.54, Bagru trading stock 82,509 units, 2,794 units in transit.
- Every mart view is served by the data API, and every dashboard tile shows its data-as-of date and its coverage status.
- At least two named business users confirm in writing that the dashboard figures match their own Excel for a sample month (sales by channel, stock at Bagru and three FBAs, open POs). Differences are logged in the issue log, not adjusted in the dashboard.
- A Stock AI run produces reorder, transfer and discontinued-stock recommendations in `insight.recommendation`; each carries its reason in plain words, the rule IDs it relied on, its inputs and any data caveat (gap IDs).
- An Operations AI run raises alerts in `insight.alert_event` for every active alert rule whose threshold is crossed, and none where it is not (verified by test cases in section 10).
- A recommendation cannot be accepted or rejected without who and when; this is enforced by the database (constraint already present).
- No customer personal data is reachable from the API or dashboards (NFR-03).

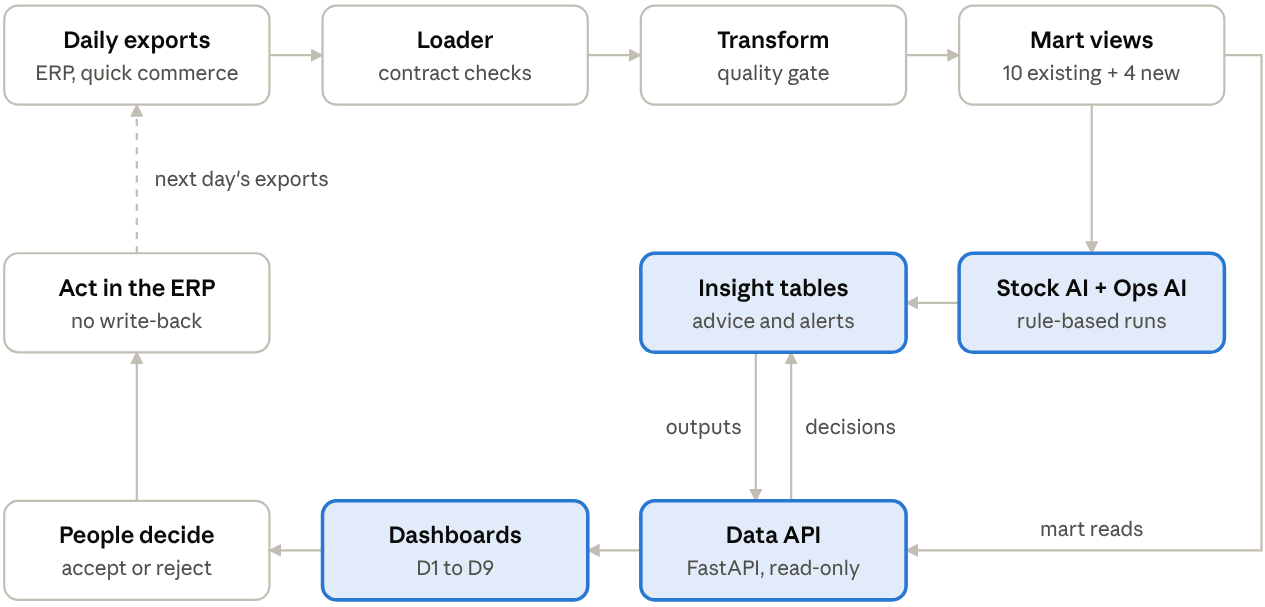
- The full 1.5-year data pull loads and builds with the quality gate passing (dependency D-01); until then Release 1 is signed off on the sample data, labelled as such.

2. Workflow

Release 1 re-uses the warehouse end to end and adds three things on top of the mart layer: a read-only data API, dashboards, and an agent runner that writes only to the `insight` schema. Nothing new writes to raw, stg or core, and the loader, transform and quality gate are unchanged.

Agents read the marts and only advise; people decide and act in the ERP

Release 1 daily cycle. Shaded boxes are new in this release.



Release 1 flow · existing pipeline, three new layers, one human decision

The agents write only advice and alerts; the only write from people is their decision on it, and the action itself happens in the ERP and returns in the next day's exports.

2.1 Daily Operating Cycle

One cycle per business day. In Release 1 every step is started by hand; scheduling is a later release.

```
1  ERP team saves the day's exports into a dated folder
   (ERP stock report, Blinkit and Instamart inventory every day; sales,
returns,
    transfers, PO report and product master as agreed per source)
        |
        v
2  python -m loader load <SOURCE> <file> [--as-of <date>]
   - contract check, all-or-nothing, audited in meta.load_batch
        |
        v
3  python -m transform run ellementry_dw
   - stg -> core -> ledger -> quality gate -> coverage -> marts, one
transaction
   - ERROR check fails  -> whole build rolled back, previous build stays,
cycle STOPS here
        |
        v
4  python -m agents run all --as-of <period_end of the latest successful
build>   <-- NEW
   - Stock AI, then Operations AI; each writes one insight.model_run
        |
        v
5  Dashboards and the data API show the new build and the new agent outputs
<-- NEW
        |
        v
6  Users review alerts and recommendations, record ACCEPTED / REJECTED with a
note <-- NEW
        |
        v
7  Users act in the ERP (raise the PO, the transfer, the markdown). Nothing
is written back.
```

2.2 Agent Run Flow

Each agent run is deterministic: the same build, settings and rule version always produce the same outputs. A run either completes and writes everything, or fails and writes nothing except its FAILED status.

```
1  AgentRunner.run(agent, as_of_date)
2  Preconditions (fail fast, no outputs):
   - latest meta.transform_run is SUCCEEDED and its period_end = as_of_date
   - mart.v_data_quality_latest has no ERROR outcome
   - every agent setting (section 6.4) is present and valid
3  INSERT insight.model_run (agent, run_type, model_name, model_version,
                             parameters = all settings used, as JSON text,
                             data_as_of_date, status = RUNNING)
4  Read mart views (core only where section 3 says so). Never raw.
5  Read meta.data_gap and mart.v_data_coverage -> which caveats apply to
which outputs
6  Compute metrics per SKU x location (section 4.2 / 5.2)
7  Apply rules -> candidate recommendations and alert events,
   each with reason (plain words), rule_ids, inputs, confidence, caveat gap
IDs
8  Supersede: earlier PROPOSED recommendations for the same type + SKU +
locations -> EXPIRED
9  Write steps 7-8 in ONE transaction; set model_run SUCCEEDED (or FAILED,
nothing written)
```

2.3 Lifecycle of an Agent Output

Output	States	Who moves it	Rule
Recommendation	PROPOSED → ACCEPTED or REJECTED (user, with who, when and an optional note); PROPOSED → EXPIRED (next run supersedes it, or it passes its expiry)	User; agent runner for EXPIRED	Accepted or rejected needs who and when (database constraint). Decided recommendations are never changed or deleted.
Alert event	OPEN → ACKNOWLEDGED → RESOLVED, or OPEN → DISMISSED	User; agent runner closes an alert as RESOLVED when a later run no longer finds the condition	Any state other than OPEN needs who and when. An alert that is still true on the next run is not raised again; its open event stays (no duplicates).

Output	States	Who moves it	Rule
Model run	RUNNING → SUCCEEDED or FAILED	Agent runner	Every output points to the run that made it, and the run stores the parameters it used.

3. Data Requirements

Both agents read the existing mart views, four new mart views and a small set of core tables; neither reads raw. Every input below exists in v2.2 except the four views marked NEW, which are added in one numbered transform step (section 8).

3.1 Inputs by Agent

Input	One row per	Stock AI uses it for	Operations AI uses it for	Today
mart.v_stock_position_latest	location × SKU, latest report	Stock on hand and sellable stock everywhere	Negative or impossible balances	Have. Snapshot history only from the first saved day (G-001). Consignment and employee balances come from the ledger with no opening balance (G-028).
mart.v_net_sales_daily	day × channel × location × SKU × partner type	Customer demand (section 3.3)	Return rate denominators	Have
mart.v_transfer_in_transit	outward transfer line not yet received	Stock already moving to a location (not recommended twice)	Transfers stuck in transit	Have

Input	One row per	Stock AI uses it for	Operations AI uses it for	Today
mart.v_open_purchase_orders	open PO line with pending quantity	Stock already on order	Overdue POs	Have. Suppliers not yet separated from return-inwards (G-005).
mart.v_supplier_lead_time	received PO line	Lead time per supplier	Supplier on-time performance	Have. Part-delivered lines approximate (G-027).
mart.v_returns_daily	day × channel × reason × location × SKU	Not used	Return volumes and spikes	Have. Reason is always "Not recorded" (G-009).
mart.v_data_coverage, mart.v_data_quality_latest, meta.data_gap	process × month; check; gap	Caveats on outputs	Data-health alerts	Have
core.fact_stock_movement	movement × location × SKU	Bagru outflow (section 3.3)	Ledger exceptions	Have. No adjustments or write-offs (G-002); goods receipts not placed (G-025).
core.fact_return_line + bridge_return_goods_receipt	return line; goods receipt	Not used	Credit-note-to-receipt lag	Have
core.fact_partner_sellout	partner × day × store × SKU	Quick-commerce consumer demand (reported)	Not used	Have for Blinkit and Instamart only (G-008)
core.fact_shipment + bridge_shipment_sku	outbound delivery	Not used	Order-to-ship time, SLA breaches, RTO by courier and pincode	Table built, 0 rows: WMS export not received (G-003)

Input	One row per	Stock AI uses it for	Operations AI uses it for	Today
core.dim_product, dim_location, dim_partner	item; place; counterparty	Lifecycle (Continued / Discontinued), item type, location type, partner type	Same	Have

3.2 New Mart Views

View (NEW)	One row per	Columns	Purpose
mart.v_stock_demand	location × SKU	location_code, location_type, sku_code, window_days, demand_units, demand_basis, avg_daily_units, days_with_demand, caveat_gap_ids	Demand per location using the definition in 3.3. Replaces the demand part of v_days_of_cover.
mart.v_network_stock	SKU	sku_code, lifecycle_status, sellable_units_by_location_type, consignment_units, in_transit_units, on_order_units, network_demand_units, network_avg_daily_units, caveat_gap_ids	Company-wide position for reorder decisions
mart.v_transfer_transit_history	received transfer line	document_number, from_location, to_location, route (from type → to type), sent_date, received_date, transit_days	Observed transit times; sets transit alert thresholds from data rather than a guess
mart.v_return_receipt_lag	domestic credit-note line	return_number, partner_type, credit_note_date, first_receipt_date, last_receipt_date, lag_days	Returns booked long before or after the goods came back (BR-023)

`mart.v_days_of_cover` is redefined on top of `v_stock_demand` (same columns, plus `demand_basis` and `caveat_gap_ids`) so dashboards built on it keep working. Its current definition divides stock by gross invoices billed at the same location; at Bagru this leaves out stock leaving through transfers and consignment, which overstates Bagru's cover.

3.3 Demand Definition by Location Type (PROPOSED, OD-02)

Demand is units, not value, counted gross before returns, over a window of `stock.demand_window_days` (PROPOSED 30) ending on the build's period end. Returns are not subtracted because returned units come back into the stock figure itself.

Location type	Demand counted	Excluded	Use in Release 1
Amazon FBA (22 locations)	Customer order lines billed at that FBA (BR-008)	—	Cover, transfer recommendations
Own store, popup store	Retail invoice lines billed at the store	—	Cover, report only
Bagru warehouse	Direct sales billed at Bagru (website, self-ship, Seller Flex, B2B, exports via Dileep) + TRANSFER_OUT to sellable locations + CONSIGNMENT_OUT to partners	Employee exhibition transfers (BR-029); ISSUE_UNCLASSIFIED (reported separately); transfers to repair or damage locations	Cover, protects stock before transfers
Blinkit and Instamart facilities	Partner sell-out units (BR-018, BR-024)	Our invoices to the partner	Cover, report only (D-092)
Franchise and SIS consignment	Net issues minus returns, as a proxy until sell-out exists (G-008)	—	Report only, flagged
Network (per SKU, for reorder)	All customer-facing demand: retail sales at every location + B2B + exports via Dileep + quick-commerce sell-out + franchise and SIS proxy	Invoices to quick-commerce partners (sell-out counts instead, never both: BR-024); transfers between our own locations; employee stock; unclassified issues	Reorder recommendations

3.4 Data Gaps That Limit Release 1

Each gap below is read from `meta.data_gap` at run time. While it is open the agent behaves as stated and names the gap ID on the affected output; when it is closed, the behaviour switches without a code change.

Gap	While open	Effect on outputs
G-001 Daily stock history	Only the latest snapshot is used	No stock trend, no ageing by date in stock
G-003 WMS dispatch export	Fulfilment metrics and alerts disabled; dashboards show "no data yet"	Operations AI covers transfers, POs, returns and data health only
G-004 Partner list (switch `partner_list_complete` = false)	Unclassified issues kept out of demand and listed	Bagru and network demand caveated; confidence at most MEDIUM where unclassified issues of that SKU exceed 10% of its demand (PROPOSED)
G-005 Factory list	Supplier on a reorder is "not identified"; lead time falls back to the all-supplier median	Reorder confidence LOW
G-007 Amazon FBA inventory reports	FBA stock is the ERP's view of FBA	Stock Amazon has not yet checked in is invisible
G-010 Set composition	A set is treated as its own SKU	Components are not depleted by set sales
G-016 Festive calendar	No seasonality	All recommendations note it from 1 Sep to 15 Nov (PROPOSED window)
G-025 Receiving location on goods receipts	Receipts stay out of the ledger	No ledger-versus-snapshot check at Bagru
G-027 Per-receipt GRN data	Lead time from the one receipt date the PO report gives	Lead times flagged approximate

Gap	While open	Effect on outputs
G-028 Opening consignment stock	Partner balances start at 1 Apr 2025 with no opening	No recall-from-partner recommendations; partner stock shown with a caveat (e.g. Blinkit consignment −5,749 units)

4. Stock AI Specification

Stock AI v1 answers four questions with fixed rules: what stock do we have where, how long will it last, what should we reorder or move, and what stock should we clear. It writes recommendations and alerts; it never changes stock or orders.

4.1 Questions Answered in Release 1

Question	Answered by	Release 1
How many sellable units of each SKU are at every location, including partners and employees?	v_stock_position_latest, dashboard	Yes, partner and employee balances caveated (G-028)
How many days will stock last at each location, and for the company as a whole?	v_days_of_cover, v_network_stock	Yes
What should we reorder, how much, and from whom?	REORDER recommendation	Yes; supplier "not identified" until G-005
What should move from Bagru to which FBA?	TRANSFER recommendation	Yes, Amazon FBA only
Which discontinued stock should we clear, and where is it?	LIQUIDATE recommendation	Yes
Which stock is not moving?	NO_RECENT_DEMAND and OVERSTOCK alerts	Yes

Question	Answered by	Release 1
Transfers to own stores and dark stores; recalls from partners	—	No (section 12)
Demand forecast	—	No; Release 1 uses the trailing average (section 12)

4.2 Metric Definitions

All metrics are per SKU, for Trading items only (BR-014), as of the build's period end. `s` = SKU, `l` = location, `W` = `stock.demand_window_days`.

Metric	Definition	Source
sellable(l, s)	sellable_qty at a location with is_sellable_stock; negative balances count as 0 and raise STK-A06	v_stock_position_latest
in_transit_to(l, s)	Units on outward transfers to l not yet received	v_transfer_in_transit
on_order(s)	Pending quantity on open PO lines (OPEN, IN_PROGRESS, OVERDUE) whose counterparty is a supplier	v_open_purchase_orders
demand(l, s)	Units per section 3.3 over W days	v_stock_demand
avg_daily(l, s)	demand(l, s) / W	v_stock_demand
cover(l, s)	(sellable + in_transit_to) / avg_daily, in days; empty when avg_daily = 0 ("no recent demand")	v_days_of_cover
network_available(s)	Sellable at all owned sellable locations + consignment balances above 0 + in transit + on order	v_network_stock

Metric	Definition	Source
lead_time(s)	Median lead time of the SKU's supplier over its last `stock.lead_time_min_receipts` receipts (PROPOSED 3). Fewer receipts or supplier unknown: the median over all suppliers in the window (23 days in the sample), flagged	v_supplier_lead_time

$$\text{reorder if } \text{network_available}(s) < \text{avg_daily}(s) \times (\text{lead_time} + \text{safety_days})$$

$$\text{reorder qty} = \left\lceil \text{avg_daily}(s) \times (\text{lead_time} + \text{review_days} + \text{safety_days}) - \text{network_available}(s) \right\rceil$$

$$\text{transfer qty}(l) = \min\left(\left\lceil \text{avg_daily}(l) \times \text{fba_target_cover} - \text{sellable}(l) - \text{in_transit_to}(l) \right\rceil,\ \text{bagru_spare} \right)$$

`bagru_spare` = Bagru sellable minus a reserve of `stock.bagru_protect_days` (PROPOSED 14) days of Bagru's direct customer demand, so website and marketplace orders shipped from Bagru are not starved by transfers.

4.3 Recommendation Rules

ID	Rule	Trigger	Quantity	Priority
FR-STK-01 REORDER	One per Continued SKU, network level, delivered to Bagru	network_available below cover for lead_time + `stock.safety_days` (PROPOSED 7)	Reorder qty formula, review period `stock.review_days` (PROPOSED 7); no rounding to pack or minimum order until those are known	HIGH if network cover < lead_time; else MEDIUM
FR-STK-02 TRANSFER	Bagru → one Amazon FBA, Continued SKU with FBA demand	cover(FBA) < `stock.fba_min_cover_days` (PROPOSED 14)	Transfer qty formula to `stock.fba_target_cover_days` (PROPOSED 30); skip if below `stock.min_transfer_units` (PROPOSED 2)	HIGH if FBA sellable = 0 or cover < 7 days (PROPOSED); else MEDIUM

ID	Rule	Trigger	Quantity	Priority
FR-STK-03 Allocation	When several FBAs need the same SKU and bagru_spare is short	—	Lowest cover first; each FBA's shortfall recorded in its reason	—
FR-STK-04 LIQUIDATE	One per Discontinued SKU (BR-015) with sellable units anywhere	Any sellable units	Total units; reason lists units and MRP value by location	HIGH if MRP value ≥ `stock.liquidate_high_value_inr` (PROPOSED Rs 50,000); else MEDIUM
FR-STK-05 Supersede	Re-running replaces, never duplicates	Same type + SKU + locations still PROPOSED	Older one set EXPIRED	—
FR-STK-06 No guessing	Unknown product, location or supplier (key −1) never receives a recommendation	—	Counted in the run notes	—

4.4 Stock Alerts

Code	Condition	Severity
STK-A01 STOCKOUT_WITH_DEMAND	Continued SKU, sellable = 0 at a sellable location with demand in the window	CRITICAL
STK-A02 LOW_COVER	cover < `stock.low_cover_days` (PROPOSED 7)	WARNING
STK-A03 OVERSTOCK	cover > `stock.overstock_cover_days` (PROPOSED 180)	INFO
STK-A04 NO_RECENT_DEMAND	Continued SKU with sellable stock and zero demand in `stock.slow_mover_days` (PROPOSED 90)	INFO
STK-A05 DISCONTINUED_ON_ORDER	Open PO pending quantity for a Discontinued SKU	WARNING
STK-A06 NEGATIVE_BALANCE	Any location × SKU balance below 0 (a missing inflow, e.g. G-028)	WARNING

4.5 Output Field Population

insight.recommendation field	REORDER	TRANSFER	LIQUIDATE
recommendation_type	REORDER	TRANSFER	LIQUIDATE
product_key	SKU	SKU	SKU
from_location_key	empty	Bagru	empty
to_location_key	Bagru	the FBA	empty
supplier_partner_key	Supplier with most receipts of the SKU, else empty	empty	empty
quantity	Reorder qty	Transfer qty	Units held
reason	Plain-words calculation, every input with its number and source	Same	Units and MRP value by location
rule_ids	BR-014, BR-015, BR-022, BR-025 + any applied	BR-008, BR-014, BR-019, BR-025	BR-015
priority	Per 4.3	Per 4.3	Per 4.3
confidence (NEW)	HIGH, MEDIUM or LOW: lowered by each caveat gap that touches the inputs	Same	HIGH unless balances are ledger-based
caveat_gap_ids (NEW)	e.g. G-005, G-027	e.g. G-007	e.g. G-028
inputs (NEW)	The metrics used, as JSON text	Same	Same
expires_on (NEW)	Run date + `insight.recommendation_expiry_days` (PROPOSED 7)	Same	Same

Example reason (REORDER): "Company-wide stock of GSTEA2724 lasts 9 days at 4.2 units a day (30-day demand). Supplier lead time is 23 days (all-supplier median: supplier not identified, G-005) plus 7 safety days. Available 38 units = 31 sellable + 7 in transit; none on order. Order 118 units to cover 23 + 7 + 7 days." Numbers here are illustrative.

5. Operations AI Specification

Operations AI v1 watches the flows that the data supports today: stock in transit between locations, purchase orders from suppliers, returns coming back, and the health of the data itself. Fulfilment (order to ship to delivery, couriers, RTO) is fully specified here but stays switched off until the WMS dispatch export arrives (G-003).

5.1 Questions Answered in Release 1

Question	Answered by	Release 1
Which transfers are in transit, for how long, and which are late for their route?	Transfers dashboard, OPS-A01	Yes
Which supplier POs are overdue, and which of them put stock at risk?	Purchasing dashboard, OPS-A02, EXPEDITE	Yes; suppliers not yet separated from return-inwards (G-005)
How reliable is each supplier on time and in full?	Supplier scorecard, OPS-A03	Partial: approximate for part deliveries (G-027)
How many units are coming back, by channel and location, and is that unusual?	Returns dashboard, OPS-A04	Yes; no return reasons (G-009)
Are partner credit notes booked far from when goods came back?	OPS-A05	Yes
Is the data fresh and the build clean?	Data health dashboard, OPS-A06 to A08	Yes
Order-to-ship time, SLA breaches, RTO by courier and pincode, stuck shipments	Fulfilment dashboard, OPS-A09 to A11	Built, switched off until G-003 closes
Warehouse throughput, courier cost	—	No (G-012; section 12)

5.2 Metric Definitions

Metric	Definition	Source
days_in_transit	Period end minus send date of an unreceived outward transfer line	v_transfer_in_transit
route_transit_p90	90th percentile of transit_days of received transfers on the same route (sending location type → receiving location), last 180 days; used only with at least `ops.route_min_observations` (PROPOSED 10)	v_transfer_transit_history
days_overdue	Period end minus due date, for supplier PO lines with pending quantity	v_open_purchase_orders
supplier_on_time_rate	Share of received PO lines whose receipt date ⩽ due date, last 180 days; part-delivered lines flagged approximate	v_supplier_lead_time (+ due date, section 8)
supplier_in_full_rate	Share of received PO lines with received_qty ⩾ order_qty	v_supplier_lead_time
weekly_return_units	Units returned per channel per ISO week (retail by return date; domestic by credit-note date, BR-023)	v_returns_daily
return_rate	Units returned ÷ units sold, same channel, trailing 8 weeks; labelled approximate because a return reverses an earlier sale	v_returns_daily, v_net_sales_daily
credit_note_lag_days	Credit-note date minus first goods-receipt date (negative = booked before goods arrived)	v_return_receipt_lag
order_to_ship_hours, ship_to_deliver_days, rto_rate	From fact_shipment timestamps and is_rto, by courier, channel and pincode	fact_shipment (G-003)

5.3 Operations Alerts

Code	Condition	Severity	Release 1
OPS-A01 TRANSIT_OVERDUE	days_in_transit > route_transit_p90; with too few observations, > `ops.transit_default_days` (PROPOSED 10)	WARNING; CRITICAL above twice the threshold	On
OPS-A02 PO_OVERDUE	Supplier PO line past due with pending quantity	WARNING; CRITICAL above `ops.po_overdue_critical_days` (PROPOSED 14)	On
OPS-A03 SUPPLIER_ON_TIME_LOW	supplier_on_time_rate < `ops.supplier_on_time_min` (PROPOSED 80%) over at least 5 receipts	INFO	On, suppliers known only after G-005
OPS-A04 RETURNS_SPIKE	Channel's weekly_return_units > trailing 8-week mean × (1 + `ops.returns_spike_pct`, PROPOSED 50%) and ≥ `ops.returns_spike_min_units` (PROPOSED 20)	WARNING	On
OPS-A05 CREDIT_NOTE_LAG	abs(credit_note_lag_days) > `ops.credit_note_lag_days` (PROPOSED 30)	INFO	On
OPS-A06 DATA_STALE	Latest stock snapshot (ERP, Blinkit, Instamart) older than 1 day on a business day, or any source older than its agreed cadence	CRITICAL for stock snapshots (a missed day cannot be re-pulled, BR-021); else WARNING	On
OPS-A07 BUILD_WARNINGS_UP	More WARN outcomes than the previous successful build	INFO	On
OPS-A08 UNCLASSIFIED_VALUE	Net sales to UNCLASSIFIED partners in the period > 0, with the value	INFO	On until G-004 closes
OPS-A09 SHIP_SLA_BREACH	Shipped after ship_by_datetime, or not shipped and past it	WARNING	Off until G-003
OPS-A10 RTO_RATE_HIGH	Courier or pincode rto_rate over 30 days > `ops.rto_rate_max` over at least 30 shipments; the threshold is set from the first 30 days of real data, not guessed	WARNING	Off until G-003
OPS-A11 SHIPMENT_STUCK	Shipped, not delivered, not RTO after `ops.delivery_max_days`	WARNING	Off until G-003

5.4 Functional Requirements

- **FR-OPS-01 EXPEDITE.** When a PO line is overdue (OPS-A02) and its SKU's network cover is below its lead time, write an EXPEDITE recommendation for that PO line, HIGH priority, naming the supplier, pending units and days overdue.

- **FR-OPS-02 Gated alerts.** An alert rule whose required process has no coverage for the period (`mart.v_data_coverage`) is skipped and logged in the run notes; its dashboard tile shows "no data yet", never 0.
- **FR-OPS-03 One open event per condition.** An alert's identity is its code + product + location (+ document for transfers and POs). If an OPEN or ACKNOWLEDGED event exists for that identity, no new event is raised; its metric value is refreshed.
- **FR-OPS-04 Auto-resolve.** An OPEN or ACKNOWLEDGED event whose condition no longer holds on the next successful run is set RESOLVED, handled_by = 'operations_ai', with the run time.
- **FR-OPS-05 Return-inwards are not supplier POs.** PO lines whose counterparty is classified as a return-inward (BR-013), or not yet classified, never feed OPS-A02, OPS-A03 or lead times; unclassified ones are counted in the run notes.
- **FR-OPS-06 Fulfilment switch-on.** When G-003 is closed and fact_shipment has rows for the period, OPS-A09 to A11 and the fulfilment dashboard turn on without a code change; thresholds marked "from data" are proposed by the first run for business sign-off and stay inactive until signed off.

6. Application: Data API, Dashboards and Settings

The application layer reads only `mart`, `insight` and the context registers in `meta`. Its single write path is a person recording a decision on a recommendation or an alert.

6.1 Data API (FastAPI)

Endpoint	Returns	Source
GET /health	Service and database status	—
GET /meta/build	Latest successful build: id, period start and end, finished at, counts of ERROR and WARN checks	meta.transform_run, v_data_quality_latest

Endpoint	Returns	Source
GET /meta/coverage, /meta/gaps, /meta/rules	Coverage by process and month; open gaps; business rules	v_data_coverage, meta.data_gap, meta.business_rule
GET /stock/positions	Stock by location × SKU; filters location, location_type, sku, lifecycle	v_stock_position_latest
GET /stock/cover, /stock/network, /stock/in-transit	Cover per location; company position per SKU; transfers in transit	v_days_of_cover, v_network_stock, v_transfer_in_transit
GET /sales/net-daily, /returns/daily	Net sales and returns; filters date range, channel, location, sku	v_net_sales_daily, v_returns_daily
GET /purchasing/open-pos, /purchasing/lead-times	Open PO lines; lead times per supplier	v_open_purchase_orders, v_supplier_lead_time
GET /insight/runs, /insight/recommendations, /insight/alerts	Agent runs and outputs; filters agent, type, status, priority, sku, location	insight.*
POST /insight/recommendations/{id}/decision	Body: decision (ACCEPTED or REJECTED), note. Who and when come from the signed-in user and the server clock	insight.recommendation
POST /insight/alerts/{id}/status	Body: status (ACKNOWLEDGED, RESOLVED, DISMISSED), note	insight.alert_event

- **FR-API-01** Every response carries `data_as_of` (build period end), `build_id` and, where the endpoint maps to a process, its coverage status for the requested months.
- **FR-API-02** Lists are paginated (default 200, maximum 1,000 rows, with a cursor). Filters are validated; an unknown filter value returns 400, not an empty list.
- **FR-API-03** Money fields are rupees excluding GST (BR-001) and named as such in the schema (e.g. `net_sales_inr_excl_gst`).
- **FR-API-04** The API connects as a database role with SELECT on mart, insight and the listed meta tables, and UPDATE only on the decision and status columns of the two insight tables. It has no access to raw or stg.
- **FR-API-05** A decision on a recommendation that is not PROPOSED, or on an expired one, is refused with 409 and changes nothing.
- **FR-API-06** An OpenAPI description is generated and kept in git; a breaking change to a response needs a new version path (/v2).

6.2 Dashboards

Release 1 dashboards are built in the tool chosen in OD-01 (PROPOSED: Metabase first, reading the same mart views, so the team can validate numbers before the custom React screens are built). The decision screens (accept or reject a recommendation, acknowledge an alert) go through the API in either case.

ID	Dashboard	Main tiles	Reads
D1	Stock overview	Units and MRP value by location type and location; Bagru vs FBA vs partners; Continued vs Discontinued	v_stock_position_latest
D2	Cover and Stock AI	Cover heatmap location × SKU; open REORDER and TRANSFER recommendations with decision buttons; stockout and low-cover alerts	v_days_of_cover, v_network_stock, insight
D3	Discontinued and slow stock	LIQUIDATE recommendations; no-demand and overstock lists	insight, v_stock_position_latest

ID	Dashboard	Main tiles	Reads
D4	Transfers	In transit by route and age; late transfers	v_transfer_in_transit, v_transfer_transit_history
D5	Purchasing and suppliers	Open and overdue POs; lead time and on-time rate per supplier; EXPEDITE recommendations	v_open_purchase_orders, v_supplier_lead_time, insight
D6	Returns	Weekly returns by channel and location; return rate; credit-note lag	v_returns_daily, v_return_receipt_lag
D7	Fulfilment	Order-to-ship, SLA breaches, RTO by courier and pincode	fact_shipment via a mart view; shows "no data yet" until G-003
D8	Data health	Latest build, quality checks, coverage by month, open gaps, unclassified partners (with value)	v_data_quality_latest, v_data_coverage, v_partner_review
D9	Sales validation	Net sales by channel and month, invoices vs returns, for reconciliation with the team's Excel	v_net_sales_daily

- **FR-DSH-01** Every tile shows its data-as-of date, and every stock tile its basis (snapshot or ledger).
- **FR-DSH-02** A process with no coverage for a period shows "no data yet" and the gap ID, never 0.
- **FR-DSH-03** Ledger-based balances (partner consignment, employees) are labelled "movements since 1 Apr 2025, no opening balance" until G-028 closes.
- **FR-DSH-04** Related-party exports (BR-007) and unclassified partners are shown as their own lines, never merged into B2B.

6.3 Access Control

Role (PROPOSED, OD-06)	Sees	Can do
Admin (data owner)	Everything in the app	Load, build, run agents, change settings, decide
Stock planner	D1 to D5, D8, D9	Accept or reject REORDER, TRANSFER, LIQUIDATE, EXPEDITE; handle stock alerts
Operations	D1, D4 to D8	Handle operations alerts; decide EXPEDITE
Management viewer	All dashboards	Read only

Release 1 runs on one laptop; users are named accounts with passwords stored hashed, and every decision records the account name. Single sign-on is a later release.

6.4 Configuration Settings

Settings live in `meta.pipeline_setting`, so each agent run can store exactly the values it used. Every value below is PROPOSED and needs business sign-off (OD-03); a run refuses to start if a setting is missing or invalid.

Setting	Proposed default	Meaning
agent.stock.enabled / agent.operations.enabled	true / true	Master switch per agent
stock.demand_window_days	30	Days of demand behind every average
stock.safety_days	7	Extra days of cover held against delay and demand swings
stock.review_days	7	How often reorders are reviewed
stock.lead_time_min_receipts	3	Receipts needed before a supplier's own lead time is used

Setting	Proposed default	Meaning
stock.fba_min_cover_days / stock.fba_target_cover_days	14 / 30	When an FBA is topped up, and to how many days
stock.bagru_protect_days	14	Bagru direct-demand days kept back from transfers
stock.min_transfer_units	2	Smallest transfer worth recommending
stock.low_cover_days / stock.overstock_cover_days	7 / 180	Alert thresholds
stock.slow_mover_days	90	No demand for this long = slow mover
stock.liquidate_high_value_inr	50000	MRP value above which a LIQUIDATE is HIGH
ops.live_mode	false	false during the fixed prototype period: freshness is judged against the build period end, not today
ops.transit_default_days / ops.route_min_observations	10 / 10	Transit threshold when a route has too little history
ops.po_overdue_critical_days	14	Overdue PO becomes CRITICAL
ops.supplier_on_time_min	0.80	Supplier on-time alert
ops.returns_spike_pct / ops.returns_spike_min_units	0.50 / 20	Returns spike alert
ops.credit_note_lag_days	30	Credit-note lag alert
insight.recommendation_expiry_days	7	PROPOSED recommendations expire after this
insight.rejection_snooze_days	7	A rejected recommendation is not re-proposed for this long unless its priority rises

Setting	Proposed default	Meaning
insight.accepted_pending_days	7	Accepted quantities count as in transit or on order until the data shows them

6.5 Non-functional Requirements

ID	Requirement
NFR-01 Reproducibility	Same build + same settings = same outputs (TC-RN-03). No randomness, no dependence on the clock except stated timestamps.
NFR-02 Performance	On the full 1.5-year data (PROPOSED targets): an agent run completes in under 5 minutes; API list endpoints answer in under 2 seconds at the 95th percentile; a dashboard loads in under 5 seconds.
NFR-03 Privacy	No customer name, phone, email or address is reachable from the API, dashboards or insight tables; analysis uses pincode, city and state only. Raw stays restricted. India's DPDP Act applies.
NFR-04 Auditability	Every output links to its run, settings, inputs and rules; every decision records who and when; nothing decided is ever changed or deleted.
NFR-05 Portability	Standard SQL by default; any PostgreSQL-only feature is listed in the README so the move to a cloud database is planned, not discovered.
NFR-06 Recoverability	Daily database backup; the whole warehouse can be rebuilt from the dated export folders.
NFR-07 Security	Least-privilege database roles; passwords stored hashed; in Release 1 the API listens on the laptop only (localhost).

7. Edge Cases and Handling

The rule for every edge case is the warehouse's own rule: never guess, count it, say it. An edge case either has a defined behaviour below or stops the run.

7.1 Data State

Edge case	Handling
The latest build failed the quality gate	Agents refuse to run (precondition, section 2.2). Dashboards keep showing the last successful build with a banner naming the failed run.
Today's stock snapshot is missing, but sales and transfers loaded	Stock AI writes no REORDER or TRANSFER for the run (stock and demand would be from different days); it still raises alerts. OPS-A06 is raised CRITICAL.
ERP, Blinkit and Instamart snapshots are from different days	Each location is read from its own system of record (BR-025) and shows its own as-of date; a gap of more than one day between them is noted on D1 and in the run notes.
A partner or employee balance is negative (e.g. Blinkit consignment −5,749 units)	Counted as 0 in every calculation; STK-A06 raised; caveat G-028 attached. Never "corrected" by an invented opening figure.
Prototype period is fixed (1 Apr 2025 to 15 Sep 2026)	The as-of date is the build's period end, not today. Setting `ops.live_mode` = false makes OPS-A06 compare against the period end, so it does not fire every day during the prototype.
A source has no data for a month (`v_data_coverage` = none)	Metrics that need it are not computed for that month; tiles show "no data yet" with the gap ID (FR-DSH-02, FR-OPS-02).

7.2 Products and Locations

Edge case	Handling
SKU in stock or sold but unknown (key −1)	No recommendation (FR-STK-06); counted on D8 with the source SKU code.
New SKU with less sales history than the window	Average uses days since its first sale or first stock, minimum 7 days (PROPOSED); reason says "new SKU, N days of history"; confidence at most MEDIUM.
Set SKUs ("Set of 2")	Treated as their own SKU; caveat G-010. Components are not depleted.
Discontinued SKU in transit to an FBA	Included in its LIQUIDATE units with the transfer document named.
Discontinued SKU still on an open PO	STK-A05; no REORDER is ever raised for a Discontinued SKU.
MRP missing for a SKU	MRP value left empty, never 0; LIQUIDATE priority MEDIUM.
Location inactive or not sellable (repair, damage, head office)	Its stock appears on D1 but is excluded from cover and recommendations.

7.3 Demand and Flows

Edge case	Handling
One large B2B or export invoice dominates a SKU's demand	Counted, but when one invoice is over 50% of the SKU's window demand (PROPOSED) the reason says so and confidence drops one level.
Unclassified partner issues (Rs 204 lakh in the sample)	Excluded from demand, listed on D8 and in OPS-A08; caveat G-004 on affected SKUs.

Edge case	Handling
Transfer received in part	The outward line counts as received today (existing view). The new transit-history view compares sent and received units; a shortfall is shown on D4 as "short received", not as stock in transit.
Two inward lines for one outward document + SKU	Handled by line_occurrence in core (existing); counted once in transit history.
Receipt dated before its PO, or lead time of 0 days	Excluded from lead times; counted as a WARN quality check.
PO line with no due date	Excluded from overdue and on-time metrics; counted on D5.
PO counterparty is a return-inward or unclassified	Never treated as a supplier (FR-OPS-05).

7.4 Decisions

Edge case	Handling
A recommendation was REJECTED and the condition persists	The same type + SKU + locations is not proposed again for `insight.rejection_snooze_days` (PROPOSED 7), unless its priority rises.
A TRANSFER or REORDER was ACCEPTED but the ERP transfer or PO is not in the data yet	Accepted quantity counts as in transit or on order for `insight.accepted_pending_days` (PROPOSED 7), so the same stock is not recommended twice; after that the data must show it.
Two users decide the same recommendation at once	The first write wins; the second gets 409 and sees the recorded decision.
A user tries to change a decided recommendation	Refused: decisions are permanent. A new decision needs a new recommendation.

8. Architectural Changes Needed

Release 1 adds three packages (agents, api, dashboards) and two numbered migrations; the loader, the raw, stg and core layers and the quality gate are untouched. The warehouse schema moves from v2.2 to v2.3.

8.1 New Components

Component	Responsibility
agents/**main**.py	CLI: `python -m agents run stock|operations|all --as-of <date> [--dry-run]`. Dry run computes and prints outputs, writes nothing.
agents/runner.py (AgentRunner)	Preconditions, model_run lifecycle, one write transaction, supersede and expiry, auto-resolve of alerts, run notes.
agents/settings.py	Reads and validates every setting from meta.pipeline_setting; returns the parameters JSON stored on the run.
agents/caveats.py	Maps open gaps (meta.data_gap) and coverage to the outputs they affect; sets caveat_gap_ids and lowers confidence.
agents/stock/ (metrics.sql, rules.py)	FR-STK-01 to 06 and STK-A01 to A06. Metrics in SQL over mart views; rules in Python; no randomness.
agents/operations/ (metrics.sql, rules.py)	FR-OPS-01 to 06 and OPS-A01 to A11.
api/ (main.py, db.py, auth.py, routers/)	FastAPI service of section 6.1; connection pool under the read-only role; decisions under the decider role.
dashboards/	Dashboard definitions exported as files and kept in git (Metabase exports, or the React app under OD-01).

Component	Responsibility
transform/sql/75_marts_agents.sql	The four new mart views; v_days_of_cover redefined on v_stock_demand; due date and on-time flag appended to v_supplier_lead_time; short-received units on transfers.
seeds/insight_alert_rule.csv	The 17 alert rules of 4.4 and 5.3 with severity, owner role and active flag; loaded by the seed loader (never deleted, only deactivated).
tests/test_agents.py, tests/test_api.py, tests/fixtures/agents/	Test cases of section 10, on small synthetic scenarios plus the sample build.

8.2 Modified Components

Component	Change
sql/110_insight_v1.sql (new migration)	insight.recommendation: add confidence (HIGH, MEDIUM, LOW), caveat_gap_ids, inputs (JSON text), expires_on, source_document (PO or transfer number). insight.alert_event: add identity_key and a unique index on identity_key for OPEN and ACKNOWLEDGED events (FR-OPS-03), plus decision_note. New schema `app` with app.user_account (name, role, password hash, active). Database roles dw_api_reader and dw_api_decider.
sql/115_agent_settings.sql (new migration)	Inserts every setting of section 6.4 with its description.
sql/990_column_definitions.sql	Regenerated: every new column gets a definition (the schema tests fail on any column without one).
seeds/column_definitions.tsv, meta_column_lineage.csv	Definitions and lineage for the new view and table columns.

Component	Change
seeds/meta_business_rule.csv	Once approved, the demand definition (OD-02) and the recommendation rules become numbered rules (BR-031 onward) and are cited in rule_ids. Until then outputs cite the existing rules only.
loader/seeds.py	Loads seeds/insight_alert_rule.csv with the same never-delete protection as the registers.
README.md, build_db.sh	New steps (agents, API, dashboards) and the two migrations.

8.3 Components Explicitly Not Modified

- **Loader, contracts and raw tables.** No change to what is accepted or how.
- **Transform steps 10 to 70 and the quality gate.** Facts, dimensions, ledger rules and ERROR checks are unchanged; step 75 runs after 70 in the same transaction.
- **Existing mart views** other than v_days_of_cover (same columns plus two) and v_supplier_lead_time (columns appended at the end). Every existing query keeps working.
- **Core tables.** No column added or changed.
- **Context registers.** Only new rows, through the existing seed CSVs.

8.4 Schema Changes

Two additive migrations (110, 115) and one regenerated file (990). No existing column is renamed, retyped or dropped, so no data migration is needed and the existing 141 tests must pass unchanged. meta.schema_version records 2.3.

9. User Stories

Roles are proposed (OD-06); the people behind them are to be named by the business.

9.1 Stock Planner

As a stock planner, I want to see sellable units of every SKU at Bagru, each FBA, our stores, the dark stores and our partners on one screen, so that I stop rebuilding the picture from separate exports.

As a stock planner, I want a list of SKUs to reorder with the quantity, the lead time used and the full calculation, so that I can check the reasoning before raising a PO.

As a stock planner, I want Bagru-to-FBA transfer suggestions that never take stock Bagru needs for its own website orders, so that topping up Amazon does not cause a website stockout.

As a stock planner, I want to accept or reject each suggestion with a note, so that the team can later see which advice was right.

9.2 Operations Lead

As an operations lead, I want to be told which transfers have been in transit longer than usual for their route, so that I can chase the right warehouse before stock is lost.

As an operations lead, I want returns by channel and week with a flag when a channel jumps, so that I can spot a packaging or listing problem early.

As an operations lead, I want order-to-ship time, SLA breaches and RTO by courier and pincode once the dispatch export arrives, so that I can hold couriers to account.

9.3 Purchasing

As the person who raises POs, I want overdue POs ranked by how much stock risk they carry, so that I chase the supplier that matters first.

As the person who raises POs, I want each supplier's real lead time and on-time rate, so that reorders use facts instead of a standard number.

9.4 Management (COO)

As the COO, I want stock value by location and the discontinued stock still held, so that I know how much money is sitting where.

As the COO, I want every dashboard to say what data it is based on and what is missing, so that I never take a decision on a zero that really means "no data".

9.5 Data Owner

As the data owner, I want an agent run to refuse to start on a failed build or missing settings, so that bad advice is never produced from bad data.

As the data owner, I want every recommendation to store the settings, inputs and rules it used, so that I can explain any output months later.

As the data owner, I want to change a threshold in one setting without touching code, so that the business can tune the rules as it learns.

9.6 ERP Team

As a member of the ERP team, I want an alert the same day a stock snapshot is missed, so that I can save it before that day's stock is lost for good.

10. Test Cases

65 test cases define when Release 1 is done. Expected outcomes are stated as things that can be checked: rows in insight tables, API responses, dashboard tiles, database refusals. Unit and integration cases run on small synthetic scenarios with known answers; end-to-end cases run on the sample build and again on the full data pull.

10.1 Baseline and Non-regression

ID	Scenario	Expected outcome	Type
TC-BL-01	Build from a clean unzip on the owner's laptop	build_db.sh, seeds and transform succeed; 78 schema, 49 loader, 6 seed and 8 transform tests pass	End-to-end
TC-BL-02	Control totals on the sample build	Retail net sales Rs 4,22,80,208.54; Bagru trading stock 82,509 units; 2,794 units in transit	End-to-end
TC-BL-03	Apply migrations 110 and 115	Applied in order; the 141 existing tests still pass; schema_version = 2.3	Integration

ID	Scenario	Expected outcome	Type
TC-BL-04	Column definitions	Every new column has a definition; schema test fails if one is removed	Integration
TC-NR-01	Transform with step 75 vs without	Same row counts and totals in every core table and existing view	Integration
TC-NR-02	Existing queries on v_days_of_cover and v_supplier_lead_time	Original columns unchanged in name, type and order	Integration

10.2 Agent Runner

ID	Scenario	Expected outcome	Type
TC-RN-01	Latest transform run FAILED	Run refused; model_run FAILED with reason; no recommendations or alerts written	Integration
TC-RN-02	A setting missing or invalid (e.g. safety_days = −1)	Run refused naming the setting; nothing written	Unit
TC-RN-03	Same build and settings run twice	Identical outputs apart from ids and timestamps; second run supersedes the first	Integration
TC-RN-04	Error raised after half the rules ran	Nothing written; model_run FAILED	Integration
TC-RN-05	Successful run	model_run.parameters holds every setting used; data_as_of_date = build period end	Unit
TC-RN-06	--dry-run	Outputs printed; insight tables unchanged	Unit

10.3 Stock Metrics

ID	Scenario	Expected outcome	Type
TC-SM-01	FBA with 60 order lines of a SKU in 30 days	demand 60, avg_daily 2.0	Unit
TC-SM-02	Bagru with direct sales, transfers out, consignment out, employee issue, unclassified issue	Demand = first three only; employee and unclassified excluded and counted	Unit
TC-SM-03	SKU invoiced to Blinkit and sold through by Blinkit	Network demand counts sell-out only, never both (BR-024)	Unit
TC-SM-04	Stock but no demand in window	cover empty, labelled "no recent demand"; never 0 or infinite	Unit
TC-SM-05	Consignment balance −20	Used as 0; STK-A06 raised with caveat G-028	Unit
TC-SM-06	Supplier with 4 receipts vs one with 2	First uses its own median; second uses the all-supplier median, flagged	Unit
TC-SM-07	Packaging item with stock	Never appears in cover or recommendations (BR-014)	Unit
TC-SM-08	New SKU first sold 10 days before period end	Average over 10 days; reason says "new SKU"; confidence ⩽ MEDIUM	Unit

10.4 Stock Recommendations

ID	Scenario	Expected outcome	Type
TC-RC-01	avg_daily 4, lead 23, safety 7, review 7, available 38	REORDER 110 units (4 × 37 − 38); HIGH (cover 9.5 < 23)	Unit
TC-RC-02	Same SKU, available 130	No REORDER	Unit
TC-RC-03	Discontinued SKU below cover	No REORDER, ever	Unit

ID	Scenario	Expected outcome	Type
TC-RC-04	FBA avg_daily 2, sellable 10, in transit 4, target 30, ample Bagru stock	TRANSFER 46 units Bagru → FBA	Unit
TC-RC-05	Bagru spare 20 after its 14-day reserve	Transfer capped at 20; shortfall stated in reason	Unit
TC-RC-06	Two FBAs short, spare covers one	Lower-cover FBA served first; the other gets the rest or nothing, with reason	Unit
TC-RC-07	FBA covered once in-transit units are counted	No TRANSFER	Unit
TC-RC-08	Computed transfer of 1 unit	Not raised (min 2)	Unit
TC-RC-09	Discontinued SKU at Bagru, an FBA and in transit	One LIQUIDATE with all units and the transfer number	Unit
TC-RC-10	Rerun with an earlier PROPOSED recommendation	Earlier one EXPIRED; one PROPOSED remains	Integration
TC-RC-11	SKU or location key −1	No recommendation; counted in run notes	Unit
TC-RC-12	Any recommendation	Reason contains every input number; rule_ids, inputs, caveat_gap_ids and expires_on are filled	Unit
TC-RC-13	Supplier unknown (G-005 open)	supplier empty; caveat G-005; confidence LOW	Unit
TC-RC-14	Rejected 3 days ago, condition persists, same priority	Not proposed again; proposed if priority rises	Integration
TC-RC-15	TRANSFER accepted yesterday, not yet in data	Accepted units count as in transit; no repeat	Integration
TC-RC-16	Snapshot older than the build period end	No REORDER or TRANSFER; OPS-A06 raised	Integration

10.5 Alerts

ID	Scenario	Expected outcome	Type
TC-AL-01	Continued SKU, sellable 0, demand in window	STK-A01 CRITICAL	Unit
TC-AL-02	cover exactly 7.0 / 6.9	Not raised / STK-A02 raised	Unit
TC-AL-03	Condition true on two runs	One OPEN event, metric refreshed	Integration
TC-AL-04	Condition gone on next run	Event RESOLVED by operations_ai with time	Integration
TC-AL-05	Route with 12 receipts, p90 = 6 days; transfer at 7 days	OPS-A01 WARNING; at 13 days CRITICAL	Unit
TC-AL-06	Route with 4 receipts; transfer at 11 days	OPS-A01 raised on the 10-day default	Unit
TC-AL-07	PO 15 days overdue	OPS-A02 CRITICAL	Unit
TC-AL-08	Return-inward or unclassified counterparty overdue	No OPS-A02; counted in notes (FR-OPS-05)	Unit
TC-AL-09	Channel week 45 units vs 8-week mean 28 / week 18 vs mean 10	Raised / not raised (below 20 units)	Unit
TC-AL-10	Credit note 40 days before first receipt	OPS-A05 INFO	Unit
TC-AL-11	live_mode = true, no ERP snapshot for yesterday	OPS-A06 CRITICAL	Unit
TC-AL-12	fact_shipment empty	OPS-A09 to A11 skipped; run notes say so; D7 shows "no data yet"	Integration
TC-AL-13	Fixture shipments loaded with breaches	OPS-A09 to A11 raised; thresholds from data stay inactive until signed off	Integration

ID	Scenario	Expected outcome	Type
TC-AL-14	Overdue PO for a SKU with network cover below lead time	EXPEDITE recommendation, HIGH, naming PO and supplier	Unit
TC-AL-15	Open PO for a Discontinued SKU	STK-A05 WARNING	Unit

10.6 Data API

ID	Scenario	Expected outcome	Type
TC-API-01	Any GET	Carries data_as_of, build_id and coverage status	Integration
TC-API-02	limit 5,000	400; default 200, maximum 1,000	Unit
TC-API-03	Unknown channel filter	400 naming the value	Unit
TC-API-04	Decision without sign-in	401; nothing changed	Integration
TC-API-05	Accept a PROPOSED recommendation	status ACCEPTED; decided_by = account; decided_at = server time	Integration
TC-API-06	Decide an EXPIRED or decided recommendation	409; nothing changed	Integration
TC-API-07	Two decisions at once	One succeeds, the other 409	Integration
TC-API-08	Reader role writes to mart or core, or reads raw	Refused by the database	Integration
TC-API-09	Scan every response schema	No customer name, phone, email or address field	Unit
TC-API-10	Net sales via API vs v_net_sales_daily	Equal to the paisa, excluding GST	Integration

10.7 Dashboards and Acceptance

ID	Scenario	Expected outcome	Type
TC-DSH-01	Month with no Blinkit data	Tile shows "no data yet" and the gap, not 0	Manual
TC-DSH-02	Every tile	Shows as-of date; stock tiles show basis	Manual
TC-DSH-03	D9 vs verified totals	Matches TC-BL-02	Manual
TC-DSH-04	Business validation	Two named users sign that D1, D5 and D9 match their Excel for one month; differences logged in the issue log	Acceptance

10.8 Test Data Requirements

Synthetic scenario fixtures, one per rule, are built as SQL inserts into a disposable database with known answers; they hold invented SKUs, locations and partners only. The sample build (Apr–Sep 2026 exports) is used for end-to-end cases, and the same end-to-end set is re-run on the full 1.5-year pull. No customer personal data appears in any fixture, test output or log.

11. Dependencies, Open Decisions and Risks

Release 1 can be built on the sample data now, but it can only be signed off for real use once the full data pull arrives and OD-02 and OD-03 are decided.

11.1 Dependencies

ID	Needed	From	Blocks	Issue / gap
D-01	Full 1.5-year data pull for every source	Business, ERP team	Sign-off on real data	D-002
D-02	Stock report saved every day (ERP, Blinkit, Instamart)	ERP team	Stock history, OPS-A06 in live mode	G-001

ID	Needed	From	Blocks	Issue / gap
D-03	Franchise, SIS and quick-commerce partner list; then `partner_list_complete` = true	Business	Accurate Bagru and network demand (Rs 204 lakh unclassified in the sample)	G-004
D-04	Factory list	Business	Suppliers on REORDER, supplier lead times, OPS-A03	G-005
D-05	WMS dispatch export as CSV, full period	Operations	OPS-A09 to A11, D7	G-003
D-06	Opening stock at sale-or-return partners on 1 Apr 2025	Business	Partner balances, recall recommendations	G-028
D-07	GRN register and receiving location on goods receipts	ERP team	Exact lead times; receipts in the ledger	G-025, G-027
D-08	Answers to the 19 open questions in the mappings workbook	Business	Correct location and channel of affected rows	Mappings v1.2
D-09	Shoppers Stop: one return route or two	Business	Returns not counted twice	D-089
D-10	Owner's laptop with PostgreSQL 16 and Python	Owner	All build work	—

11.2 Open Decisions

ID	Decision	Proposal	Needed before
OD-01	First dashboards: Metabase or the custom React app	Metabase first to validate numbers with the team; React for the decision screens and later the full app	Epic E-3
OD-02	Demand definition by location type (section 3.3)	As proposed; then recorded as a business rule	Epic E-4

ID	Decision	Proposal	Needed before
OD-03	Every PROPOSED setting (section 6.4) and who owns each	Values as proposed, owned by the stock planner and operations lead	Go-live
OD-04	Release 1 transfers to Amazon FBA only	Yes; own stores and dark stores in a later release	Epic E-4
OD-05	Alert delivery outside the app	In-app only in Release 1; email or WhatsApp in Release 2	Release 2
OD-06	Roles and the named people in each	As in section 6.3	Epic E-6
OD-07	Language model provider, and whether Release 2 adds plain-language explanations and questions over the insight tables	Decide after Release 1 acceptance data exists	Release 2
OD-08	Approver of this PRD	To be named by the business	Sign-off
OD-09	Do factories ever deliver straight to an FBA or store, or always to Bagru?	Always Bagru (REORDER delivers to Bagru)	Epic E-4
OD-10	Count exports via Dileep (at cost, BR-007) in network demand?	Yes: they consume the same stock; shown separately in the reason	Epic E-4

11.3 Product Risks

Risk	Likelihood / impact	Mitigation
Simple rules give poor advice for some SKUs (lumpy B2B demand, festive peaks)	High / Medium	Every recommendation needs a human decision; acceptance rate and reasons tracked by SKU type; forecasting in Release 3 once a full year exists

Risk	Likelihood / impact	Mitigation
Sample data is not representative of the full period	Medium / High	Sign-off only on the full pull (D-01); end-to-end tests re-run on it
Unclassified partners distort demand	High / Medium (until D-03)	Excluded and caveated; value shown on D8 every day
The team does not trust the numbers and keeps its own Excel	Medium / High	D9 reconciliation and written sign-off (TC-DSH-04) before agents are used
Daily snapshots are missed and history is lost	Medium / High	OPS-A06 CRITICAL the same day; ERP team owns the daily save
One person builds and runs everything	High / Medium	Every step scripted and tested; README run book; no step depends on memory

12. Future Enhancements (Out of Current Scope)

Each item below waits for data, for evidence from Release 1, or both. None needs a redesign: each adds views, rules or a component on the same warehouse and insight schema.

12.1 Plain-language Explanations and Questions (Release 2)

A language model reads the insight tables, the mart views and the rule register, and answers questions such as "why should we reorder this?" or "what is stuck in transit?" in plain words, citing rule and gap IDs. It never computes numbers itself; it quotes the agents' stored calculations. Needs OD-07 and LangGraph orchestration.

12.2 Demand Forecasting (Release 3)

Replace the trailing average with a forecast per SKU, location and week, stored in `insight.demand_forecast` with its range and model version. Needs at least one full year including the festive season (D-01), the festive and sale-event calendar (G-016), and Release 1's acceptance data to compare against.

12.3 More Transfer Destinations

Transfers to own stores, and replenishment of Blinkit and Instamart dark stores once their supply route is settled (D-092). Recall-from-partner recommendations once opening partner stock exists (G-028).

12.4 Set and Component Planning

Deplete component stock when sets sell, and reorder components rather than sets, once set composition is supplied (G-010).

12.5 Alert Delivery and Scheduling

Daily scheduled load, build and agent runs (n8n for collection, Python for logic, per the architecture), with alerts sent to email, WhatsApp or Slack (OD-05).

12.6 Fuller Operations Picture

Amazon FBA inventory and inbound reports (G-007), marketplace return reasons (G-009), courier cost per shipment (G-012) and warehouse throughput, giving cost per order and true FBA availability.

12.7 Value-based Planning

Gross margin and cash tied up per recommendation once product cost is available (G-021), so reorders and markdowns can be ranked by money, not only by units.

Annexure A. Engineering Task Breakdown

50 tasks in 8 epics, 174 story points, indicative only. Each task names the PRD section it implements and the test cases that must pass for it to be done. No dates are set: the work is done by one person and paced by the dependencies in section 11.

A.1 Conventions

Convention	Description
Hierarchy	Epic (E-#) > Task (AIH-###)

Convention	Description
Story points	1 trivial · 2 one file with tests · 3 small new component · 5 component with an integration · 8 cross-component · 13 must be split
Definition of Ready	Requirement and test cases referenced; settings it needs exist; dependencies done
Definition of Done	Code in git; its test cases green; all 141 existing tests green; README and column definitions updated; no PROPOSED value hard-coded

A.2 E-1 Laptop Baseline

Task	Description	Pts	Refs
AIH-001	Install PostgreSQL 16 and the pinned Python environment; build from a clean unzip	2	§1.3; TC-BL-01
AIH-002	Reproduce the control totals; record them in the run book	1	TC-BL-02
AIH-003	Run book in README: load, build, run agents, back up, restore	2	§2.1
AIH-004	Daily database backup and dated export folder layout	2	D-02; NFR-06

A.3 E-2 Agent Views and Migrations

Task	Description	Pts	Refs
AIH-005	Migration 110: insight v1 columns, alert identity index, app schema, database roles	5	§8.2; TC-BL-03, TC-API-08
AIH-006	Migration 115: settings, with validation rules	2	§6.4; TC-RN-02
AIH-007	v_stock_demand per location type	8	§3.3; TC-SM-01 to 03, 07, 08

Task	Description	Pts	Refs
AIH-008	v_network_stock	5	§3.2; TC-SM-03, TC-RC-01
AIH-009	v_transfer_transit_history with short-received units	3	§5.2; TC-AL-05, 06
AIH-010	v_return_receipt_lag; due date on v_supplier_lead_time; v_days_of_cover on v_stock_demand	3	§3.2; TC-AL-10, TC-NR-02
AIH-011	Column definitions and lineage for every new column	2	TC-BL-04

A.4 E-3 Data API

Task	Description	Pts	Refs
AIH-012	FastAPI skeleton, connection pool, configuration, /health	3	§6.1
AIH-013	Response envelope (data_as_of, build_id, coverage); /meta endpoints	3	FR-API-01; TC-API-01
AIH-014	Stock, sales, returns and purchasing endpoints; pagination; filter validation	5	FR-API-02, 03; TC-API-02, 03, 10
AIH-015	Insight read endpoints	2	§6.1
AIH-016	Decision and status endpoints; 409 rules; concurrent decisions	5	FR-API-05; TC-API-05 to 07
AIH-017	Accounts, password hashing, sign-in, roles	5	§6.3; TC-API-04
AIH-018	Personal-data scan of response schemas; OpenAPI file in git	2	FR-API-06; TC-API-09

A.5 E-4 Stock AI

Task	Description	Pts	Refs
AIH-019	AgentRunner: preconditions, model_run, one transaction, dry run	5	§2.2; TC-RN-01 to 06
AIH-020	Caveat engine from meta.data_gap and coverage	3	§3.4; TC-RC-13
AIH-021	Stock metrics SQL	5	§4.2; TC-SM-01 to 08
AIH-022	REORDER rule	5	FR-STK-01; TC-RC-01 to 03
AIH-023	TRANSFER rule, Bagru reserve, allocation	8	FR-STK-02, 03; TC-RC-04 to 08
AIH-024	LIQUIDATE rule	2	FR-STK-04; TC-RC-09
AIH-025	Supersede, expiry, rejection snooze, accepted-pending	5	FR-STK-05; §7.4; TC-RC-10, 14, 15
AIH-026	Reason text and inputs JSON	3	§4.5; TC-RC-12
AIH-027	Stock alerts STK-A01 to A06	3	§4.4; TC-AL-01, 02, 15; TC-SM-05

A.6 E-5 Operations AI

Task	Description	Pts	Refs
AIH-028	Alert identity, no duplicates, auto-resolve	3	FR-OPS-03, 04; TC-AL-03, 04
AIH-029	OPS-A01 transit overdue	3	TC-AL-05, 06
AIH-030	OPS-A02, A03; return-inward exclusion	3	FR-OPS-05; TC-AL-07, 08

Task	Description	Pts	Refs
AIH-031	OPS-A04, A05	3	TC-AL-09, 10
AIH-032	OPS-A06 to A08 data health; live mode	3	TC-AL-11
AIH-033	EXPEDITE recommendation	2	FR-OPS-01; TC-AL-14
AIH-034	Fulfilment alerts A09 to A11 behind the coverage switch	5	FR-OPS-02, 06; TC-AL-12, 13
AIH-035	Alert rule seed file and seed-loader protection	2	§8.1

A.7 E-6 Dashboards and Decisions

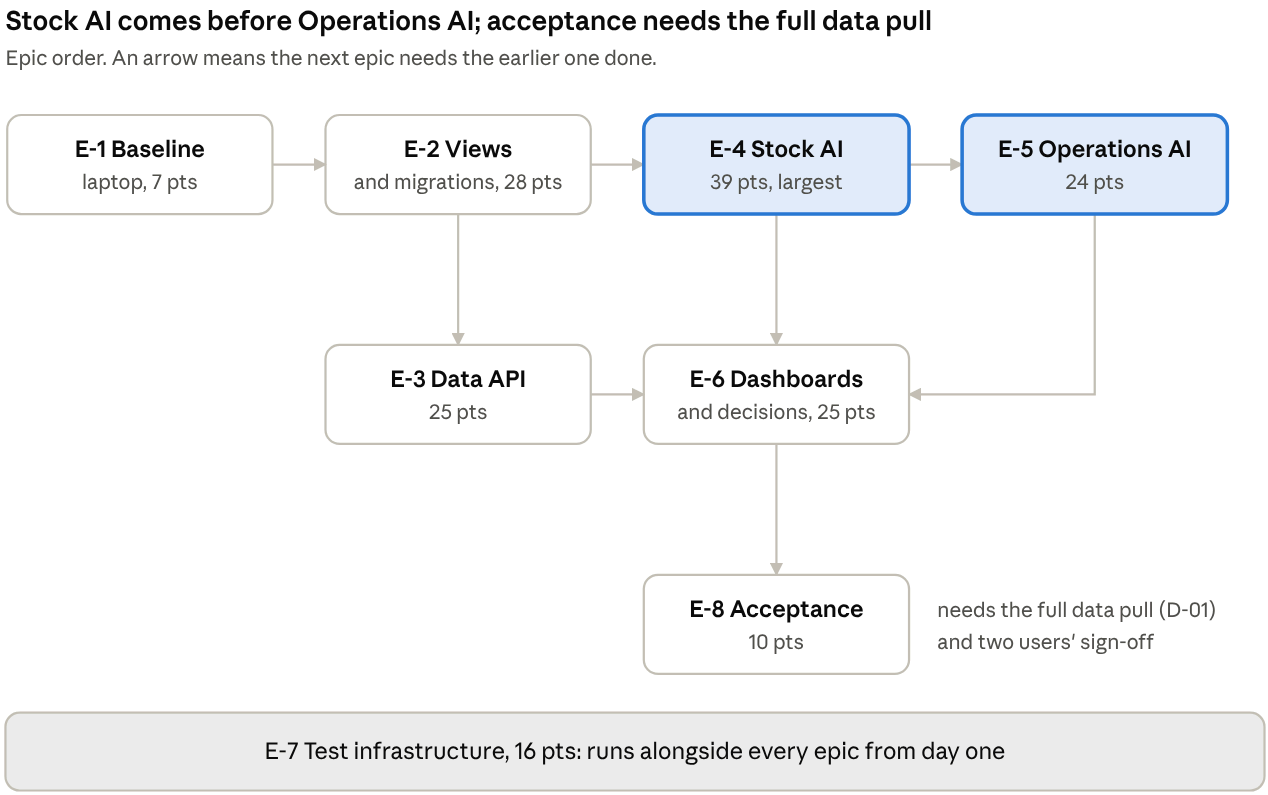
Task	Description	Pts	Refs
AIH-036	Dashboard tool on the read-only role (per OD-01)	2	OD-01
AIH-037	D1 Stock overview, D2 Cover and Stock AI	5	§6.2; TC-DSH-02
AIH-038	D3 Discontinued, D4 Transfers, D5 Purchasing	5	§6.2
AIH-039	D6 Returns, D7 Fulfilment (no-data state), D8 Data health	5	FR-DSH-02; TC-DSH-01
AIH-040	D9 Sales validation	2	TC-DSH-03
AIH-041	Decision screens through the API	5	§2.3
AIH-042	Dashboard definitions exported to git	1	§8.1

A.8 E-7 Test Infrastructure and E-8 Acceptance

Task	Description	Pts	Refs
AIH-043	Scenario fixture framework: disposable database, inserts, known answers	5	§10.8

Task	Description	Pts	Refs
AIH-044	Stock rule fixtures	3	§10.3, 10.4
AIH-045	Operations rule fixtures, including synthetic shipments	3	§10.5
AIH-046	Non-regression harness: totals before and after step 75	3	TC-NR-01
AIH-047	One-command test run	2	Definition of Done
AIH-048	Business validation sessions and written sign-off	3	TC-DSH-04
AIH-049	Full data pull: load, build, end-to-end re-run	5	D-01
AIH-050	Threshold sign-off; approved rules recorded as BR-031 onward	2	OD-02, OD-03

A.9 Dependency Map



epic dependency map · 8 epics

Operations AI follows Stock AI because the EXPEDITE rule reuses the agent runner and the stock metrics; the API can be built alongside Stock AI once the views exist.

A.10 Aggregate Sizing

Epic	Tasks	Points	Note
E-1 Laptop baseline	4	7	First; nothing else starts before it
E-2 Views and migrations	7	28	Demand view is the hardest single piece
E-3 Data API	7	25	Can run alongside E-4
E-4 Stock AI	9	39	Largest; transfer allocation is the riskiest rule
E-5 Operations AI	8	24	Fulfilment part waits on G-003 for real data only
E-6 Dashboards and decisions	7	25	Shape depends on OD-01
E-7 Test infrastructure	5	16	Parallel from day one
E-8 Acceptance	3	10	Needs D-01
Total	**50**	**174**	Indicative; refine as work starts

A.11 Engineering Risk Register

Risk	Likelihood / impact	Mitigation
Demand view double counts (invoices plus sell-out, transfers plus sales)	Medium / High	TC-SM-02 and 03; a reconciliation check that network demand equals the sum of its parts by source
Views too slow on 1.5 years of data	Medium / Medium	Measure against NFR-02 on the full pull; materialise heavy views as tables in step 75 if needed
The dashboard tool cannot write decisions	High / Low	By design decisions go through the API screens (AIH-041)

Risk	Likelihood / impact	Mitigation
Laptop failure loses the database	Low / High	Daily backup (AIH-004); every build is reproducible from the dated export folders
PostgreSQL-only SQL creeps in	Medium / Low	Postgres-only features listed in the README; portable SQL by default (NFR-05)
Settings changed between runs make outputs hard to compare	Medium / Medium	Parameters stored on every run; dashboards show the run's settings